Zhang, Dongliang (张东良)

CONTACT INFORMATION

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EDUCATION

Peking University, China 2015- 2020

Ph.D., Biophysics and Condensed Matter Physics, 2020
- Thesis title: Nonequilibrium thermodynamics for information processing in living systems
- Mentor: Qi Ouyang and Yuhai Tu

Peking University, China 2011–2015

B.A., Physics, 2015
B.A., Psychology (double degree), 2015

RESEARCH EXPERIENCE

Max Planck Institute for the Physics of Complex Systems 2024 – present

Postdoctoral associate, working with **Frank Jülicher**.
- Energetic cost for spindle assembly
- Energy-induced pattern switch among spatially-extended coupled oscillators

Tsinghua University 2023 – 2024

Research associate at Department of Physics working with **Yuansheng Cao**, **Qi Ouyang** (PKU) and **Yuhai Tu** (IBM)
- Mechanism of synchronization among active agents
- Thermodynamic mechanism of nonlinear traveling wave

Peking University 2020 – 2023

Postdoctoral fellow at School of Physics working with **Qi Ouyang** (PKU) and **Yuhai Tu** (IBM)
- Mechanism of synchronization among cyanobacteria circadian clocks
- Revealing subunit cooperativity in cyanobacteria circadian protein hexamer with statistics of cryo-EM structural data
- Nonequilibrium thermodynamics for accurate positioning of Turing pattern in finite systems

Peking University 2015 – 2020

Graduate (Ph.D.) student mentored by **Yuhai Tu** (IBM), **Qi Ouyang** (PKU)
- The free energy cost for synchronization of coupled molecular oscillators
- Synchronization through monomer shuffling among cyanobacteria circadian clocks
- Scaling law of energy dissipation rate in nonequilibrium reaction networks

Harvard University Oct. 2019 – Jan. 2020

Short-term visiting student, hosted by **Ying Lu** (Harvard Medical School)
- Nonequilibrium thermodynamics for cooperativity in proteasome ATPase complex

Peking University 2014 – 2015

Undergraduate research. Advisor: **Qi Ouyang** (PKU)
- Reverse engineering of functional circuits by combining Boolean network model and continuous model

Peking University 2013 – 2015

Undergraduate research. Advisor: **Haitao Quan** (PKU), **Biao Wu** (PKU)
- Ergodicity and mixing in quantum dynamics

TEACHING AND MENTORING

Teaching assistant of Nonlinear Physics, PKU 2018

Mentored undergraduate students (Shengyao Luo, Chenghao Zhang, Qiwei Yu, Sayun Mao, Jingye Wang), PKU 2018–2024

SKILLS

Stochastic processes and stochastic simulations
Nonlinear dynamics, nonequilibrium thermodynamics, and statistical physics
Biological modeling and simulation

C and MATLAB programming Vibe coding with LLM

HONORS AND AWARDS

Boya Postdoctoral fellowship, Peking Univerisity	2020
Quantitative Biology Research Scholarship, Peking University	2019
Headmaster Scholarship, Peking University	2019
Award for Scientific Research, Peking University	2016
DTZ Scholarship, Peking University	2016

PUBLICATIONS

1. **Dongliang Zhang**, Yuansheng Cao, Qi Ouyang, Yuhai Tu (2025). Altruistic resource-sharing mechanism for synchronization: The energy-speed-accuracy tradeoff. Physical Review Letters, 135(3), 037401.
2. Xu Han, **Dongliang Zhang**, Lu Hong, Daqi Yu, Zhaolong Wu, Tian Yang, Michael Rust, Yuhai Tu, Qi Ouyang (2023). A cooperative switch within the KaiC hexamer revealed by cryo-EM. Nature Communications, 14, 5907.
3. **Dongliang Zhang**[†], Chenghao Zhang[†], Qi Ouyang, Yuhai Tu (2023). Free energy dissipation enhances spatial accuracy and robustness of self-positioned Turing pattern in small biochemical systems. Journal of the Royal Society Interface, 20(204), 20230276. ([†]co-first authors)
4. Qiwei Yu, **Dongliang Zhang**, Yuhai Tu (2021). Inverse power law scaling of energy dissipation rate in nonequilibrium reaction networks. Physical Review Letters, 126(8), 080601.
5. **Dongliang Zhang**, Qi Ouyang. (2021). Nonequilibrium thermodynamics in biochemical systems and its application. Entropy, 23(3), 271.
6. **Dongliang Zhang**, Yuansheng Cao, Qi Ouyang, Yuhai Tu (2020). The energy cost and optimal design for synchronization of coupled molecular oscillators. Nature physics, 16(1) : 95-100.
7. **Dongliang Zhang**, H. T. Quan, Biao Wu (2016). Ergodicity and mixing in quantum dynamics. Physical Review E, 94(2), 022150.
8. Bin Shao, Xiang Liu, **Dongliang Zhang**, Jiayi Wu, Qi Ouyang (2015). From Boolean network model to continuous model helps in design of functional circuits. PloS one, 10(6), e0128630.